

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING  
AND TECHNOLOGY

**BASIC MECHANICAL ENGINEERING**

**MODEL QUESTION PAPER – SET-1**

Time: 3 hours

Maximum Marks: 75

**PART A**

**I. Answer all the following questions in one word or sentence.**

**(9 x 1 = 9 Marks)**

|   |                                                                       | Module Outcome | Cognitive level |
|---|-----------------------------------------------------------------------|----------------|-----------------|
| 1 | Ability of a material to absorb energy is called .....                | <b>M1.01</b>   | U               |
| 2 | Cochran boiler is a ..... Boiler.                                     | <b>M1.02</b>   | R               |
| 3 | List out any two boiler accessories.                                  | <b>M2.03</b>   | R               |
| 4 | ..... is a positive displacement compressor.                          | <b>M2.04</b>   | R               |
| 5 | What is the unit of refrigeration?                                    | <b>M3.02</b>   | R               |
| 6 | Efficiency of a refrigerator is expressed in .....                    | <b>M3.04</b>   | R               |
| 7 | Which gear is used for power transmission of two intersecting shafts? | <b>M3.04</b>   | R               |
| 8 | ..... is a solid lubricant.                                           | <b>M4.02</b>   | R               |
| 9 | Point out the types of belt drive.                                    | <b>M4.04</b>   | R               |

**PART B**

**II. Answer any Eight questions from the following**

**(8 x 3 = 24 Marks)**

|   |                                                                                      | Module Outcome | Cognitive level |
|---|--------------------------------------------------------------------------------------|----------------|-----------------|
| 1 | List any three mechanical properties.                                                | <b>M1.01</b>   | R               |
| 2 | Point out any 3 properties of a good lubricant.                                      | <b>M4.03</b>   | U               |
| 3 | What is the classification of boilers?                                               | <b>M3.02</b>   | U               |
| 4 | Draw the schematic diagram of a reciprocating compressor and mark the salient parts. | <b>M3.02</b>   | U               |
| 5 | Define ton of refrigeration.                                                         | <b>M3.02</b>   | R               |

|    |                                               |              |   |
|----|-----------------------------------------------|--------------|---|
| 6  | What do you mean by COP of a refrigerator?    | <b>M3.03</b> | R |
| 7  | Differentiate between Spur and Helical gears. | <b>M4.04</b> | U |
| 8  | Classify bearings.                            | <b>M4.04</b> | R |
| 9  | Differentiate open and cross belt drives.     | <b>M4.05</b> | U |
| 10 | Draw Splash lubricating system.               | <b>M4.02</b> | U |

### PART C

#### III. Answer all questions from the following

**(6 x 7 = 42 Marks)**

Module Outcome Cognitive level

|    |                                                                         |              |   |
|----|-------------------------------------------------------------------------|--------------|---|
| 1  | Describe various types of Cast Iron                                     | <b>M1.04</b> | R |
| OR |                                                                         |              |   |
| 2  | Explain any 5 mechanical properties.                                    | <b>M1.03</b> | U |
| 3  | What are the merits and demerits of a solid fuel?                       | <b>M1.04</b> | U |
| OR |                                                                         |              |   |
| 4  | Explain properties of a good fuel.                                      | <b>M1.02</b> | U |
| 5  | Explain the working of centrifugal compressors.                         | <b>M2.03</b> | U |
| OR |                                                                         |              |   |
| 6  | What are the various types of refrigerants?                             | <b>M2.03</b> | R |
| 7  | Explain the properties of a good refrigerant.                           | <b>M2.03</b> | U |
| OR |                                                                         |              |   |
| 8  | What are the various types of gears used in gear transmissions?         | <b>M2.04</b> | U |
| 9  | What are the basic components of a hydraulic power transmission system? | <b>M3.04</b> | U |
| OR |                                                                         |              |   |
| 10 | What are the types of lubricants?                                       | <b>M3.04</b> | R |

|    |                                                 |              |          |
|----|-------------------------------------------------|--------------|----------|
| 11 | What are the components of a pneumatic system?  | <b>M4.02</b> | <b>U</b> |
| OR |                                                 |              |          |
| 12 | What are the merits and demerits of gear drive? | <b>M4.03</b> | <b>U</b> |

Prepared By:

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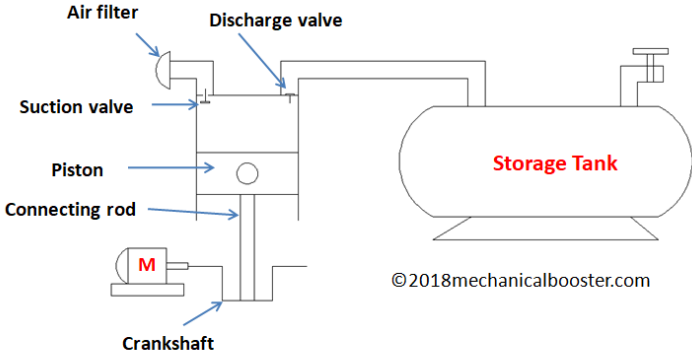
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## SCHEME OF VALUATION – SET I

### (SCORING INDICATORS)

| <div style="display: flex; justify-content: space-between;"> <span><b>Revision : 2021</b></span> <span><b>Course Code :</b></span> </div> <div style="display: flex; justify-content: space-between;"> <span><b>Course Title : Basic Mechanical Engineering</b></span> </div> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                               |                      |              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|----------------------|--------------|
| <b>Q.N<br/>O</b>                                                                                                                                                                                                                                                              | <b>SCORING INDICATORS</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <b>SPLIT<br/>UP<br/>SCORE</b> | <b>SUB<br/>TOTAL</b> | <b>TOTAL</b> |
| I                                                                                                                                                                                                                                                                             | <b>PART A</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                               |                      |              |
| 1.                                                                                                                                                                                                                                                                            | Strength                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                             | 1                    | 1            |
| 2.                                                                                                                                                                                                                                                                            | Fire tube boiler                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 1                             | 1                    | 1            |
| 3.                                                                                                                                                                                                                                                                            | Air preheater, Economiser,                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 0.5*2                         | 1                    | 1            |
| 4.                                                                                                                                                                                                                                                                            | Reciprocating compressor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                             | 1                    | 1            |
| 5.                                                                                                                                                                                                                                                                            | Ton of refrigeration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1                             | 1                    | 1            |
| 6.                                                                                                                                                                                                                                                                            | Coefficient of performance (C O P)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1                             | 1                    | 1            |
| 7.                                                                                                                                                                                                                                                                            | Bevel gears                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1                             | 1                    | 1            |
| 8                                                                                                                                                                                                                                                                             | Graphite                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1                             | 1                    | 1            |
| 9                                                                                                                                                                                                                                                                             | Open belt drive ,Cross belt drive                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 0.5*2                         | 1                    | 1            |
| II                                                                                                                                                                                                                                                                            | <b>PART B</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                               |                      |              |
| 1.                                                                                                                                                                                                                                                                            | Hardness, Toughness, Ductility<br><br>Out of 10 mechanical properties,list out any 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1*3                           | 3                    | 3            |
| 2.                                                                                                                                                                                                                                                                            | <ul style="list-style-type: none"> <li>A high boiling point and low freezing point (in order to stay liquid within a wide range of temperature)</li> <li>A high viscosity index.</li> <li>Thermal stability.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3                             | 3                    | 3            |
| 3.                                                                                                                                                                                                                                                                            | <p style="text-align: center;"><i><b>Classification of Boiler</b></i></p> <pre> graph TD     Root[Classification of Boiler] --&gt; Nature[Nature]     Root --&gt; PosWater[Position of water]     Root --&gt; Axis[Axis]     Root --&gt; PosFurnace[Position of furnace]     Root --&gt; Circulation[Circulation]     Root --&gt; Fuel[Fuel]     Root --&gt; NoTube[No. of tube]          Nature --&gt; Stationary[Stationary]     Nature --&gt; Marine[Marine Locomotive]          PosWater --&gt; WaterTube[Water Tube]     PosWater --&gt; FireTube[Fire Tube]          Axis --&gt; Vertical[Vertical]     Axis --&gt; Horizontal[Horizontal]          PosFurnace --&gt; Internally[Internally]     PosFurnace --&gt; Externally[Externally]          Circulation --&gt; Free[Free]     Circulation --&gt; Forced[Forced]          Fuel --&gt; Solid[Solid]     Fuel --&gt; Liquid[Liquid]     Fuel --&gt; Gas[Gas]     Fuel --&gt; Nuclear[Nuclear]          NoTube --&gt; Single[Single]     NoTube --&gt; Multitubular[Multitubular] </pre> | 3                             | 3                    | 3            |

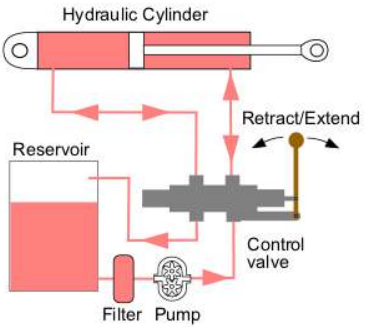
|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |   |   |   |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| 4. |  <p style="text-align: center;"><b>Main Parts of Reciprocating Air Compressor</b></p>                                                                                                                                                                                                                                                                                                                                                                                                   | 3 | 3 | 3 |
| 5. | <ul style="list-style-type: none"> <li>• Ton of Refrigeration(TR) is the unit of refrigeration capacity.</li> <li>• 1 TR indicates the amount of heat that needs to be extracted from 1 tonne of water at zero degree centigrade to convert it to 1 tonne of ice at zero degree centigrade in 1 day.</li> <li>• 1 TR = 3.5 KW = 210 KJ/min</li> </ul>                                                                                                                                                                                                                     | 3 | 3 | 3 |
| 6. | <p>The <b>coefficient of performance, COP</b>, of a <b>refrigerator</b> is defined as the heat removed from the cold reservoir <math>Q_{\text{cold}}</math>, (i.e. inside a refrigerator) divided by the work <math>W</math> done to remove the heat (i.e. the work done by the compressor).</p>                                                                                                                                                                                                                                                                          | 3 | 3 | 3 |
| 7. | <p>In spur gear, the teeth are parallel to the axis of the gear. In helical gear, teeth are inclined at an angle (called helix angle) with the gear axis.</p> <p>Spur gear imposes only radial load on bearings. Helical gear imposes radial load and axial thrust load on bearings.</p> <p>Radial bearings that can handle radial load only can be employed here. Bearings that can handle both radial and axial load must be employed here.</p> <p>Spur gear can be used for parallel shafts only. Helical gears can be used for parallel shafts or crossed shafts.</p> | 3 | 3 | 3 |

|        |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |   |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| 8      |                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3 | 3 | 3 |
| 9      | <ul style="list-style-type: none"> <li>• two pulleys rotate in the same direction</li> <li>• Length of the belt is smaller</li> <li>• Angle of lap is different for driver and driven pulley</li> </ul> <ul style="list-style-type: none"> <li>• pulleys rotate in the opposite directions</li> <li>• Length of the belt is larger</li> <li>• Angle of lap is same for driver and driven pulley</li> </ul>                                         | 3 | 3 | 3 |
| 10     |                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3 | 3 | 3 |
| PART C |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |   |
| 1.     | <p>There are four basic types of cast iron</p> <ul style="list-style-type: none"> <li>• white iron</li> <li>• gray iron</li> <li>• ductile iron</li> <li>• malleable iron</li> </ul> <p><b>White Cast Iron</b></p> <p>Characterized by the prevalence of carbides, impacting</p> <ul style="list-style-type: none"> <li>• high compressive strength</li> <li>• hardness</li> <li>• good resistance to wear</li> </ul> <p><b>Gray Cast Iron</b></p> | 7 | 7 | 7 |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |   |   |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
|    | <p>Characterized with graphite in the microstructure, giving</p> <ul style="list-style-type: none"> <li>• good machinability</li> <li>• good resistance to wear and galling</li> </ul> <p><b>Ductile Cast Iron</b></p> <p>Gray iron with small amounts of magnesium and cesium which nodulates the graphite, resulting</p> <ul style="list-style-type: none"> <li>• high strength</li> <li>• high ductility</li> </ul> <p><b>Malleable Cast Iron</b></p> <p>White cast iron heat-treated to improve</p> <ul style="list-style-type: none"> <li>• higher ductility</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                        |   |   |   |
|    | Or                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |   |   |
| 2. | Strength ,Malleability,Hardness,Toughness,Elasticity definitions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 7 | 7 | 7 |
|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |   |   |
| 3. | <p><b>Advantages of solid fuels :</b></p> <ul style="list-style-type: none"> <li>• They are easy to transport.</li> <li>• They are convenient to store without any risk of spontaneous explosion.</li> <li>• Their cost of production is low.</li> <li>• They possess moderate ignition temperature.</li> </ul> <p><b>Disadvantages of solid fuels :</b></p> <ul style="list-style-type: none"> <li>• Their ash content is high.</li> <li>• They burn with clinker formation.</li> <li>• They combustion operations cannot be controlled easily.</li> <li>• Their cost of handling is high.</li> <li>• Their large proportion of heat is wasted during combustion.</li> <li>• Their thermal efficiency is low.</li> <li>• Their <a href="#">calorific value</a> is lower as compared to that of liquids fuels.</li> <li>• They require a large excess of air for complete combustion.</li> <li>• They cannot be used as internal combustion engine fuels</li> </ul> | 7 | 7 | 7 |
|    | or                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |   |   |
| 4. | <ul style="list-style-type: none"> <li>• It should be easily available. ...</li> <li>• It should be cheap.</li> <li>• It should have a high calorific value.</li> <li>• It should have a low ignition point, which is not lower than room temperature (so that it does not catch fire at room temperature).</li> <li>• It should not burn too fast or too slowly.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 7 | 7 | 7 |
|    | OR                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |   |   |

| 5.                          | The Principle of Centrifugal Compression:<br><br>1. The air reaches the center of the impeller.<br>2. Air is forced outward by centrifugal force.<br>3. Diffuser gradually reduces the air velocity.<br>4. Velocity energy is converted to higher pressure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                     |                                 |  | 7      | 7                      | 7                                 |                                 |                        |                        |                               |             |                           |                                             |              |           |                          |                                                                     |                                    |                    |                            |                                     |                      |         |                             |                                                                      |                                     |                        |                      |                                 |        |           |   |   |   |
|-----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|---------------------------------|--|--------|------------------------|-----------------------------------|---------------------------------|------------------------|------------------------|-------------------------------|-------------|---------------------------|---------------------------------------------|--------------|-----------|--------------------------|---------------------------------------------------------------------|------------------------------------|--------------------|----------------------------|-------------------------------------|----------------------|---------|-----------------------------|----------------------------------------------------------------------|-------------------------------------|------------------------|----------------------|---------------------------------|--------|-----------|---|---|---|
|                             | or                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                     |                                 |  |        |                        |                                   |                                 |                        |                        |                               |             |                           |                                             |              |           |                          |                                                                     |                                    |                    |                            |                                     |                      |         |                             |                                                                      |                                     |                        |                      |                                 |        |           |   |   |   |
| 6.                          | <table><thead><tr><th>Sector</th><th>Type of equipment used</th><th>Refrigerants currently being used</th><th>Typical refrigerant charge (kg)</th></tr></thead><tbody><tr><td>Domestic Refrigeration</td><td>Domestic refrigerators</td><td>R-134a Hydrocarbons (limited)</td><td>0.05 – 0.25</td></tr><tr><td>Domestic Air Conditioning</td><td>Window and split type air conditioning unit</td><td>R-22, R-134a</td><td>0.5 – 1.5</td></tr><tr><td>Commercial Refrigeration</td><td>Display cabinets and refrigerated or chilled furniture. Cold Rooms:</td><td>R-134a, R-502, R-404A, R408C, R-22</td><td>0.2 – 5<br/>15 - 25</td></tr><tr><td>Industrial Food Processing</td><td>Large chest freezers and cold rooms</td><td>R-502, R-22, Ammonia</td><td>5 - 250</td></tr><tr><td>Commercial air Conditioning</td><td>- Large split type air conditioning units<br/>- Water chiller systems</td><td>R-22, R-407A, R-134a, R 410a, R407c</td><td>1.2 – 2.0<br/>125 – 270</td></tr><tr><td>Vehicle Conditioning</td><td>Vehicle air conditioning system</td><td>R-134a</td><td>0.8 – 1.0</td></tr></tbody></table> |                                     |                                 |  | Sector | Type of equipment used | Refrigerants currently being used | Typical refrigerant charge (kg) | Domestic Refrigeration | Domestic refrigerators | R-134a Hydrocarbons (limited) | 0.05 – 0.25 | Domestic Air Conditioning | Window and split type air conditioning unit | R-22, R-134a | 0.5 – 1.5 | Commercial Refrigeration | Display cabinets and refrigerated or chilled furniture. Cold Rooms: | R-134a, R-502, R-404A, R408C, R-22 | 0.2 – 5<br>15 - 25 | Industrial Food Processing | Large chest freezers and cold rooms | R-502, R-22, Ammonia | 5 - 250 | Commercial air Conditioning | - Large split type air conditioning units<br>- Water chiller systems | R-22, R-407A, R-134a, R 410a, R407c | 1.2 – 2.0<br>125 – 270 | Vehicle Conditioning | Vehicle air conditioning system | R-134a | 0.8 – 1.0 | 7 | 7 | 7 |
| Sector                      | Type of equipment used                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Refrigerants currently being used   | Typical refrigerant charge (kg) |  |        |                        |                                   |                                 |                        |                        |                               |             |                           |                                             |              |           |                          |                                                                     |                                    |                    |                            |                                     |                      |         |                             |                                                                      |                                     |                        |                      |                                 |        |           |   |   |   |
| Domestic Refrigeration      | Domestic refrigerators                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | R-134a Hydrocarbons (limited)       | 0.05 – 0.25                     |  |        |                        |                                   |                                 |                        |                        |                               |             |                           |                                             |              |           |                          |                                                                     |                                    |                    |                            |                                     |                      |         |                             |                                                                      |                                     |                        |                      |                                 |        |           |   |   |   |
| Domestic Air Conditioning   | Window and split type air conditioning unit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | R-22, R-134a                        | 0.5 – 1.5                       |  |        |                        |                                   |                                 |                        |                        |                               |             |                           |                                             |              |           |                          |                                                                     |                                    |                    |                            |                                     |                      |         |                             |                                                                      |                                     |                        |                      |                                 |        |           |   |   |   |
| Commercial Refrigeration    | Display cabinets and refrigerated or chilled furniture. Cold Rooms:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | R-134a, R-502, R-404A, R408C, R-22  | 0.2 – 5<br>15 - 25              |  |        |                        |                                   |                                 |                        |                        |                               |             |                           |                                             |              |           |                          |                                                                     |                                    |                    |                            |                                     |                      |         |                             |                                                                      |                                     |                        |                      |                                 |        |           |   |   |   |
| Industrial Food Processing  | Large chest freezers and cold rooms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | R-502, R-22, Ammonia                | 5 - 250                         |  |        |                        |                                   |                                 |                        |                        |                               |             |                           |                                             |              |           |                          |                                                                     |                                    |                    |                            |                                     |                      |         |                             |                                                                      |                                     |                        |                      |                                 |        |           |   |   |   |
| Commercial air Conditioning | - Large split type air conditioning units<br>- Water chiller systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | R-22, R-407A, R-134a, R 410a, R407c | 1.2 – 2.0<br>125 – 270          |  |        |                        |                                   |                                 |                        |                        |                               |             |                           |                                             |              |           |                          |                                                                     |                                    |                    |                            |                                     |                      |         |                             |                                                                      |                                     |                        |                      |                                 |        |           |   |   |   |
| Vehicle Conditioning        | Vehicle air conditioning system                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | R-134a                              | 0.8 – 1.0                       |  |        |                        |                                   |                                 |                        |                        |                               |             |                           |                                             |              |           |                          |                                                                     |                                    |                    |                            |                                     |                      |         |                             |                                                                      |                                     |                        |                      |                                 |        |           |   |   |   |
|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     |                                 |  |        |                        |                                   |                                 |                        |                        |                               |             |                           |                                             |              |           |                          |                                                                     |                                    |                    |                            |                                     |                      |         |                             |                                                                      |                                     |                        |                      |                                 |        |           |   |   |   |
| 7.                          | <ul style="list-style-type: none"><li>• Low Freezing Point. Refrigerants should have low freezing point than the normal operating conditions. ...</li><li>• Low Condensing Pressure. ...</li><li>• High Evaporator Pressure. ...</li><li>• High Critical Pressure. ...</li><li>• High Vapor Density. ...</li><li>• High Dielectric strength. ...</li><li>• High Latent Heat of Vaporization. ...</li><li>• High Heat Transfer Coefficient</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                     |                                 |  | 7      | 7                      | 7                                 |                                 |                        |                        |                               |             |                           |                                             |              |           |                          |                                                                     |                                    |                    |                            |                                     |                      |         |                             |                                                                      |                                     |                        |                      |                                 |        |           |   |   |   |
|                             | or                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                     |                                 |  |        |                        |                                   |                                 |                        |                        |                               |             |                           |                                             |              |           |                          |                                                                     |                                    |                    |                            |                                     |                      |         |                             |                                                                      |                                     |                        |                      |                                 |        |           |   |   |   |
| 8.                          | Spur Gear.<br>Helical Gear. ...<br>Double Helical Gear. ...<br>Herringbone Gear. ...<br>Bevel Gear. ...<br>Worm Gear. ...<br>Hypoid Gear.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                     |                                 |  | 7      | 7                      | 7                                 |                                 |                        |                        |                               |             |                           |                                             |              |           |                          |                                                                     |                                    |                    |                            |                                     |                      |         |                             |                                                                      |                                     |                        |                      |                                 |        |           |   |   |   |
|                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     |                                 |  |        |                        |                                   |                                 |                        |                        |                               |             |                           |                                             |              |           |                          |                                                                     |                                    |                    |                            |                                     |                      |         |                             |                                                                      |                                     |                        |                      |                                 |        |           |   |   |   |
| 9.                          | Selector valve, hydraulic lines, actuating units, reservoir, and                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                     |                                 |  |        | 7                      | 7                                 |                                 |                        |                        |                               |             |                           |                                             |              |           |                          |                                                                     |                                    |                    |                            |                                     |                      |         |                             |                                                                      |                                     |                        |                      |                                 |        |           |   |   |   |



|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |   |   |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
|     | <p><b>pump.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 7 |   |   |
|     | or                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |   |   |
| 10. | There are 4 types of lubricants: <b>Oil, Grease, Penetrating Lubricants, and Dry Lubricants</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 7 | 7 | 7 |
|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |   |   |   |
| 11. | <ul style="list-style-type: none"> <li>• Check Valves. ...</li> <li>• Compressor. ...</li> <li>• Regulators and gauges. ...</li> <li>• Accumulator or buffer tank. ...</li> <li>• Feed lines. ...</li> <li>• Actuators</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 7 | 7 | 7 |
|     | or                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |   |   |
| 12. | <p><b>Advantages</b></p> <ol style="list-style-type: none"> <li>1. By using gear trains, large <u>velocity ratio</u> can be obtained with minimum space.</li> <li>2. Gears are mechanically strong, so higher loads can be lifted.</li> <li>3. Gears are used for transmission of large H.F.</li> <li>4. They are used for transmitting motion over small centre distance of shafts</li> <li>5. They are used for large reduction in speed and for transmission of torque.</li> <li>6. Gears require only lubrication, hence less maintenance is required.</li> <li>7. Using gear systems, we can transmit motion between non-parallel intersecting shafts.</li> <li>8. They are used for positive drive, so its <u>velocity</u> ratio remains constant.</li> <li>9. They have long life, so the gear system is very compact.</li> </ol> <p><b>Disadvantages</b></p> <ol style="list-style-type: none"> <li>1. They are not suitable for large <u>velocities</u>.</li> <li>2. They are not suitable for transmitting motion over a large distance.</li> <li>3. Due to the engagement of toothed wheel of gears, some part of machine may get permanently damaged in case of excessive loading.</li> <li>4. They have no flexibility.</li> <li>5. Gear operation is noisy</li> </ol> | 7 | 7 | 7 |

# BLUE PRINT

## Mark Distribution

| Module | Hr/ Module                        | Marks/ Module<br>( $h_i / \sum H_i$ ) * 123 ( $\pm 5\%$ ) | Type of Questions |       |                  |       |                  |       |                  |       |
|--------|-----------------------------------|-----------------------------------------------------------|-------------------|-------|------------------|-------|------------------|-------|------------------|-------|
|        |                                   |                                                           | Part A            |       | Part B           |       | Part C           |       | Total            |       |
|        |                                   |                                                           | No. of Questions  | Marks | No. of Questions | Marks | No. of Questions | Marks | No. of Questions | Marks |
| 1      | 11                                | 31.5                                                      | 2                 | 2     | 1                | 3     | 4                | 28    | 7                | 33    |
| 2      | 12                                | 34.3                                                      | 3                 | 3     | 1                | 3     | 4                | 28    | 8                | 34    |
| 3      | 10                                | 28.6                                                      | 2                 | 2     | 4                | 12    | 2                | 14    | 8                | 28    |
| 4      | 10                                | 28.6                                                      | 2                 | 2     | 4                | 12    | 2                | 14    | 8                | 28    |
| Total  | 43(+2<br>for<br>internal<br>exam) | 123.0                                                     | 9                 | 9     | 10               | 30    | 12               | 84    | 31               | 123   |

## BLUE PRINT

### COGNITIVE LEVEL MARK DISTRIBUTION[Cognitive wise question analysis]

| Cognitive Level | Marks | % of Marks |
|-----------------|-------|------------|
| Remembering     | 42    | 34         |
| Understanding   | 81    | 66         |
| Applying        | 0     | 0          |
| Analysing       | 0     | 0          |
| Evaluating      | 0     | 0          |
| Creating        | 0     | 0          |
| Total           | 123   | 100        |

## QUESTION WISE ANALYSIS

**Course: Basic Mechanical Engineering**

| <b>Q. No.</b> | <b>Module Outcome</b> | <b>Cognitive Level</b> | <b>Marks</b> | <b>Time</b> |
|---------------|-----------------------|------------------------|--------------|-------------|
| <b>I.1</b>    | <b>M1.01</b>          | <b>Understanding</b>   | <b>1</b>     | <b>2</b>    |
| <b>I.2</b>    | <b>M1.02</b>          | <b>Remembering</b>     | <b>1</b>     | <b>2</b>    |
| <b>I.3</b>    | <b>M2.03</b>          | <b>Remembering</b>     | <b>1</b>     | <b>2</b>    |
| <b>I.4</b>    | <b>M2.04</b>          | <b>Remembering</b>     | <b>1</b>     | <b>2</b>    |
| <b>I.5</b>    | <b>M3.02</b>          | <b>Remembering</b>     | <b>1</b>     | <b>2</b>    |
| <b>I.6</b>    | <b>M3.04</b>          | <b>Understanding</b>   | <b>1</b>     | <b>2</b>    |
| <b>I.7</b>    | <b>M3.04</b>          | <b>Remembering</b>     | <b>1</b>     | <b>2</b>    |
| <b>I.8</b>    | <b>M4.02</b>          | <b>Remembering</b>     | <b>1</b>     | <b>2</b>    |
| <b>I.9</b>    | <b>M4.04</b>          | <b>Remembering</b>     | <b>1</b>     | <b>2</b>    |
| <b>II.1</b>   | <b>M1.01</b>          | <b>Remembering</b>     | <b>3</b>     | <b>7.5</b>  |
| <b>II.2</b>   | <b>M4.03</b>          | <b>Understanding</b>   | <b>3</b>     | <b>7.5</b>  |
| <b>II.3</b>   | <b>M3.02</b>          | <b>Understanding</b>   | <b>3</b>     | <b>7.5</b>  |
| <b>II.4</b>   | <b>M3.02</b>          | <b>Understanding</b>   | <b>3</b>     | <b>7.5</b>  |
| <b>II.5</b>   | <b>M3.02</b>          | <b>Remembering</b>     | <b>3</b>     | <b>7.5</b>  |
| <b>II.6</b>   | <b>M3.03</b>          | <b>Remembering</b>     | <b>3</b>     | <b>7.5</b>  |
| <b>II.7</b>   | <b>M4.04</b>          | <b>Understanding</b>   | <b>3</b>     | <b>7.5</b>  |
| <b>II.8</b>   | <b>M4.04</b>          | <b>Remembering</b>     | <b>3</b>     | <b>7.5</b>  |
| <b>II.9</b>   | <b>M4.05</b>          | <b>Understanding</b>   | <b>3</b>     | <b>7.5</b>  |
| <b>II.10</b>  | <b>M4.02</b>          | <b>Understanding</b>   | <b>3</b>     | <b>7.5</b>  |
| <b>III</b>    | <b>M1.04</b>          | <b>Remembering</b>     | <b>7</b>     | <b>17</b>   |
| <b>IV</b>     | <b>M1.03</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>V</b>      | <b>M1.04</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>VI</b>     | <b>M1.02</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>VII</b>    | <b>M2.03</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>VIII</b>   | <b>M2.03</b>          | <b>Remembering</b>     | <b>7</b>     | <b>17</b>   |
| <b>IX</b>     | <b>M2.03</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>X</b>      | <b>M2.04</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>XI</b>     | <b>M3.04</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>XII</b>    | <b>M3.04</b>          | <b>Remembering</b>     | <b>7</b>     | <b>17</b>   |
| <b>XIII</b>   | <b>M4.02</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>XIV</b>    | <b>M4.03</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |

THIRD SEMESTER DIPLOMA EXAMINATION IN ENGINEERING  
AND TECHNOLOGY

**BASIC MECHANICAL ENGINEERING  
MODEL QUESTION PAPER – SET-II**

Time: 3 hours

Maximum Marks: 75

**PART A**

**IV. Answer all the following questions in one word or sentence.**

**(9 x 1 = 9 Marks)**

|   |                                                                         | Module Outcome | Cognitive level |
|---|-------------------------------------------------------------------------|----------------|-----------------|
| 1 | Which mechanical property needed for a material to resist impact loads? | <b>M1.04</b>   | <b>R</b>        |
| 2 | Write an example of water tube boiler.                                  | <b>M1.02</b>   | <b>R</b>        |
| 3 | Fusible plug is a .....                                                 | <b>M2.03</b>   | <b>R</b>        |
| 4 | Quality of steam can be determined by the term .....                    | <b>M2.04</b>   | <b>R</b>        |
| 5 | One ton of refrigeration is ..... kW.                                   | <b>M3.02</b>   | <b>R</b>        |
| 6 | A good lubricant has high ..... and low .....                           | <b>M3.04</b>   | <b>R</b>        |
| 7 | Velocity ratio is the ratio between speed of ..... and .....            | <b>M3.04</b>   | <b>R</b>        |
| 8 | Refrigerant used in domestic refrigeration is .....                     | <b>M4.02</b>   | <b>R</b>        |
| 9 | Roller bearings are also called .....                                   | <b>M4.04</b>   | <b>R</b>        |

**PART B**

**V. Answer any Eight questions from the following**

**(8 x 3 = 24 Marks)**

|   |                                                     | Module Outcome | Cognitive level |
|---|-----------------------------------------------------|----------------|-----------------|
| 1 | Draw the stress-strain curve of a ductile material. | <b>M1.01</b>   | <b>R</b>        |
| 2 | What are the various types of steel?                | <b>M2.04</b>   | <b>U</b>        |
| 3 | List out the merits of water tube boilers.          | <b>M3.02</b>   | <b>U</b>        |
| 4 | Draw the diagram of a rotary compressor             | <b>M3.02</b>   | <b>U</b>        |
| 5 | What is a refrigerant?                              | <b>M3.02</b>   | <b>R</b>        |

|    |                                                             |              |          |
|----|-------------------------------------------------------------|--------------|----------|
| 6  | What are the components of a refrigerator?                  | <b>M3.03</b> | <b>R</b> |
| 7  | What are the applications of bevel gears?                   | <b>M4.04</b> | <b>U</b> |
| 8  | Differentiate between sliding and rolling contact bearings. | <b>M4.04</b> | <b>R</b> |
| 9  | List out the advantages of belt drives.                     | <b>M4.05</b> | <b>U</b> |
| 10 | Point out the merits of a lubricating system.               | <b>M4.02</b> | <b>U</b> |

### PART C

#### VI. Answer all questions from the following (6 x 7 = 42 Marks)

Module Outcome    Cognitive level

|    |                                                                |              |          |
|----|----------------------------------------------------------------|--------------|----------|
| 1  | What are the various types of steel based on carbon content?   | <b>M1.04</b> | <b>R</b> |
| OR |                                                                |              |          |
| 2  | Discuss the alloying elements used in steel.                   | <b>M1.03</b> | <b>U</b> |
| 3  | What are the merits and demerits of gaseous fuels?             | <b>M1.04</b> | <b>U</b> |
| OR |                                                                |              |          |
| 4  | Discuss calorific value in detail.                             | <b>M1.02</b> | <b>U</b> |
| 5  | Explain the stages in a reciprocating compressor.              | <b>M2.03</b> | <b>U</b> |
| OR |                                                                |              |          |
| 6  | Explain Vapour compression refrigeration system with sketches. | <b>M2.03</b> | <b>R</b> |
| 7  | Discuss the classification of refrigerants.                    | <b>M2.03</b> | <b>U</b> |
| OR |                                                                |              |          |
| 8  | What are the functions of each element in a refrigerator?      | <b>M2.04</b> | <b>U</b> |
| 9  | Describe the advantages of pneumatic systems.                  | <b>M3.04</b> | <b>U</b> |
| OR |                                                                |              |          |
| 10 | Explain the merits and demerits of a hydraulic system.         | <b>M3.04</b> | <b>R</b> |

|    |                                                     |              |          |
|----|-----------------------------------------------------|--------------|----------|
| 11 | What are the properties of a good bearing material? | <b>M4.03</b> | <b>U</b> |
| OR |                                                     |              |          |
| 12 | Explain the pressurised lubricating system.         | <b>M4.03</b> | <b>U</b> |

Prepared By:

Dhanya V

Lecturer

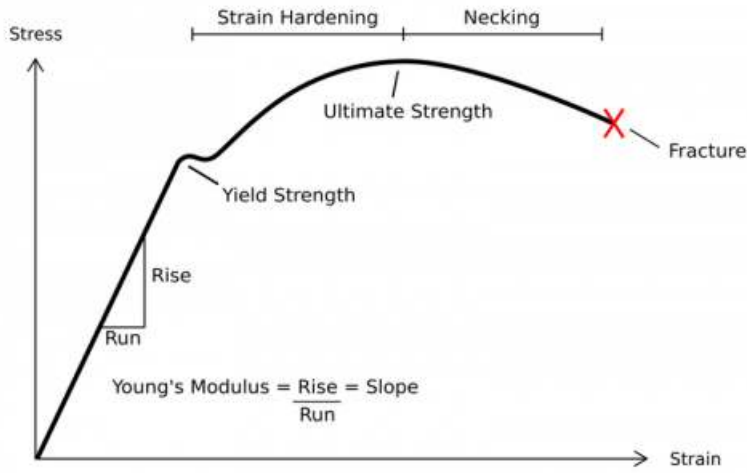
Department of Polymer Technology

GPTC Adoor

9946517840

## SCHEME OF VALUATION- SET II

### (SCORING INDICATORS)

| Revision : 2021                             |                                                                                      | Course Code :             |                  |           |
|---------------------------------------------|--------------------------------------------------------------------------------------|---------------------------|------------------|-----------|
| Course Title : Basic Mechanical Engineering |                                                                                      |                           |                  |           |
| Q.N<br>O                                    | SCORING INDICATORS                                                                   | SPLI<br>T UP<br>SCO<br>RE | SUB<br>TOT<br>AL | TOT<br>AL |
| I                                           | PART A                                                                               |                           |                  |           |
| 1.                                          | Toughness                                                                            | 1                         | 1                | 1         |
| 2.                                          | La Mont boiler                                                                       | 1                         | 1                | 1         |
| 3.                                          | Boiler mounting                                                                      | 1                         | 1                | 1         |
| 4.                                          | Dryness fraction                                                                     | 1                         | 1                | 1         |
| 5.                                          | 3.5kW                                                                                | 1                         | 1                | 1         |
| 6.                                          | Boiling point, freezing point                                                        | 0.5*2                     | 1                | 1         |
| 7.                                          | Driver, driven                                                                       | 0.5*2                     | 1                | 1         |
| 8                                           | Freon-12                                                                             | 1                         | 1                | 1         |
| 9                                           | Anti-friction bearing                                                                | 1                         | 1                | 1         |
| II                                          | PART B                                                                               |                           |                  |           |
| 1.                                          |  | 1*3                       | 3                | 3         |

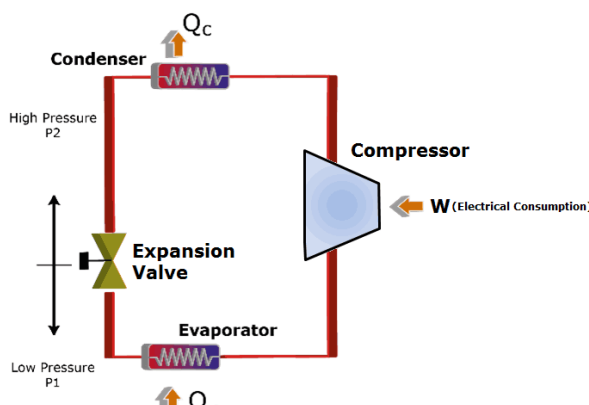


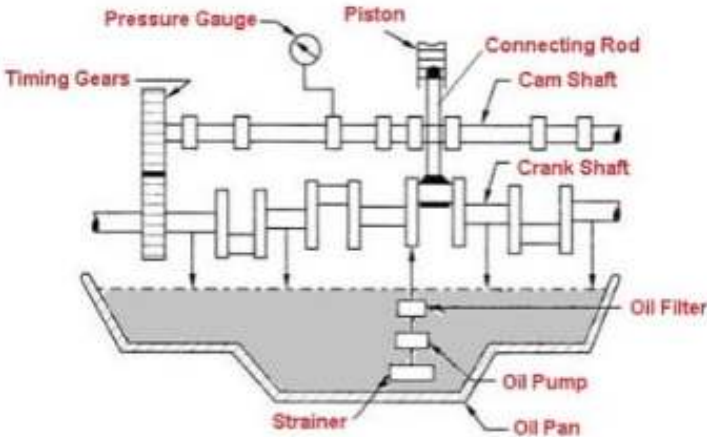
|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |   |   |   |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| 2. | <pre> graph LR     Steel --&gt; Carbon_Steel[Carbon Steel]     Steel --&gt; Alloy_Steel[Alloy Steel]     Steel --&gt; Stainless_Steel[Stainless Steel]     Steel --&gt; Tool_Steel[Tool Steel]     Carbon_Steel --&gt; Low_Carbon_Steel[Low Carbon Steel]     Carbon_Steel --&gt; Medium_Carbon_Steel[Medium Carbon Steel]     Carbon_Steel --&gt; High_Carbon_Steel[High Carbon Steel]     Alloy_Steel --&gt; Ferritic[Ferritic]     Alloy_Steel --&gt; Martensitic[Martensitic]     Alloy_Steel --&gt; Austenitic[Austenitic]     Stainless_Steel --&gt; Duplex[Duplex]     Stainless_Steel --&gt; PH[Precipitation Hardening(PH)] </pre> | 3 | 3 | 3 |
| 3. | <p style="text-align: center;"><b><u>Water-tube Boilers</u></b></p> <p><b>Advantages:</b></p> <ul style="list-style-type: none"> <li>• Small water content</li> <li>• Small diameter tubes and steam drum</li> <li>• Able to produce superheated steam</li> <li>• Customizable design (flexibility)</li> </ul>                                                                                                                                                                                                                                                                                                                              | 3 | 3 | 3 |
| 4. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3 | 3 | 3 |
| 5. | A refrigerant is a <b>working fluid used in the refrigeration cycle of air conditioning systems</b> and heat pumps where in most cases they undergo a repeated phase transition from a liquid to a gas and back again.                                                                                                                                                                                                                                                                                                                                                                                                                      | 3 | 3 | 3 |
| 6. | <p><b>The 4 Main Refrigeration Cycle Components</b></p> <p>The compressor.</p> <p>The condenser.</p> <p>The expansion device.</p> <p>The evaporator.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 3 | 3 | 3 |
| 7. | The bevel gear has many diverse applications such as <b>locomotives, marine applications, automobiles, printing presses, cooling towers, power plants, steel plants, railway track inspection machines, etc</b>                                                                                                                                                                                                                                                                                                                                                                                                                             | 3 | 3 | 3 |

| 8      | <table><tr><th>SR.NO</th><th>Parameter</th><th>Sliding contact bearing</th><th>Rolling contact bearing</th></tr><tr><td>1</td><td>Size</td><td>large</td><td>Small</td></tr><tr><td>2</td><td>starting torque</td><td>High</td><td>low</td></tr><tr><td>3</td><td>noise</td><td>Less noise</td><td>High noise</td></tr><tr><td>4</td><td>Life</td><td>Less life</td><td>Long life</td></tr><tr><td>5</td><td>Cost</td><td>Less cost</td><td>More costly</td></tr><tr><td>6</td><td>Coeff. of friction</td><td>High</td><td>less</td></tr></table> | SR.NO                   | Parameter               | Sliding contact bearing | Rolling contact bearing | 1 | Size | large | Small | 2 | starting torque | High | low | 3 | noise | Less noise | High noise | 4 | Life | Less life | Long life | 5 | Cost | Less cost | More costly | 6 | Coeff. of friction | High | less | 3 | 3 | 3 |
|--------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-------------------------|-------------------------|-------------------------|---|------|-------|-------|---|-----------------|------|-----|---|-------|------------|------------|---|------|-----------|-----------|---|------|-----------|-------------|---|--------------------|------|------|---|---|---|
| SR.NO  | Parameter                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Sliding contact bearing | Rolling contact bearing |                         |                         |   |      |       |       |   |                 |      |     |   |       |            |            |   |      |           |           |   |      |           |             |   |                    |      |      |   |   |   |
| 1      | Size                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | large                   | Small                   |                         |                         |   |      |       |       |   |                 |      |     |   |       |            |            |   |      |           |           |   |      |           |             |   |                    |      |      |   |   |   |
| 2      | starting torque                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | High                    | low                     |                         |                         |   |      |       |       |   |                 |      |     |   |       |            |            |   |      |           |           |   |      |           |             |   |                    |      |      |   |   |   |
| 3      | noise                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Less noise              | High noise              |                         |                         |   |      |       |       |   |                 |      |     |   |       |            |            |   |      |           |           |   |      |           |             |   |                    |      |      |   |   |   |
| 4      | Life                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Less life               | Long life               |                         |                         |   |      |       |       |   |                 |      |     |   |       |            |            |   |      |           |           |   |      |           |             |   |                    |      |      |   |   |   |
| 5      | Cost                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Less cost               | More costly             |                         |                         |   |      |       |       |   |                 |      |     |   |       |            |            |   |      |           |           |   |      |           |             |   |                    |      |      |   |   |   |
| 6      | Coeff. of friction                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | High                    | less                    |                         |                         |   |      |       |       |   |                 |      |     |   |       |            |            |   |      |           |           |   |      |           |             |   |                    |      |      |   |   |   |
| 9      | <ul style="list-style-type: none"><li>• Belt drives are cost-effective. ...</li><li>• They are simple to use.</li><li>• Belt drives do not require parallel shaft.</li><li>• They have a low maintenance cost.</li><li>• They come with overload and jam protection.</li><li>• Different speeds can be obtained by means of step or tapered pulleys.</li></ul>                                                                                                                                                                                    | 3                       | 3                       | 3                       |                         |   |      |       |       |   |                 |      |     |   |       |            |            |   |      |           |           |   |      |           |             |   |                    |      |      |   |   |   |
| 10     | <ul style="list-style-type: none"><li>• Long-term protection of equipment.</li><li>• Increased mechanical reliability.</li><li>• Cartridge change in less than 60 seconds!</li><li>• Accurate lubrication.</li><li>• Reduced maintenance costs.</li><li>• Increased profitability.</li><li>• The right lubricant volume per point.</li></ul>                                                                                                                                                                                                      | 3                       | 3                       | 3                       |                         |   |      |       |       |   |                 |      |     |   |       |            |            |   |      |           |           |   |      |           |             |   |                    |      |      |   |   |   |
| PART C |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                         |                         |                         |                         |   |      |       |       |   |                 |      |     |   |       |            |            |   |      |           |           |   |      |           |             |   |                    |      |      |   |   |   |
| 1.     | <p><b>Steel</b></p> <ul style="list-style-type: none"><li>* Mild steel<ul style="list-style-type: none"><li>* Carbon content - 0.25%</li></ul></li><li>* Low alloy<ul style="list-style-type: none"><li>* Sulphur - 0.035%</li><li>* Phosphorus - 0.035%</li></ul></li><li>* Medium carbon steel<ul style="list-style-type: none"><li>* Carbon - 0.25% - 0.6%</li></ul></li><li>* High carbon steel or hard steel<ul style="list-style-type: none"><li>* Carbon - 0.6% - 1.0%</li></ul></li></ul>                                                 | 7                       | 7                       | 7                       |                         |   |      |       |       |   |                 |      |     |   |       |            |            |   |      |           |           |   |      |           |             |   |                    |      |      |   |   |   |
|        | Or                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                         |                         |                         |                         |   |      |       |       |   |                 |      |     |   |       |            |            |   |      |           |           |   |      |           |             |   |                    |      |      |   |   |   |

| 2.                                          | <table><tr><th>Alloying Element</th><th>Effect on Steel</th></tr><tr><td>Aluminum</td><td>Deoxidation – ease of nitriding</td></tr><tr><td>Boron 2-10 ppm, How added (effective Boron)</td><td>Hardenability</td></tr><tr><td>Carbon</td><td>Hardness – strength – wear</td></tr><tr><td>Chromium</td><td>Corrosion resistance - hardenability</td></tr><tr><td>Cobalt aka</td><td>Hardness – wear Reduces hardenability</td></tr><tr><td>Columbium (Niobium)</td><td>Elimination of carbide precipitation</td></tr><tr><td>Copper</td><td>Corrosion resistance – strength</td></tr><tr><td>Lead Not used anymore</td><td>Machinability</td></tr><tr><td>Manganese</td><td>Strength – hardenability – more response to heat treatment</td></tr><tr><td>Molybdenum</td><td>High temperature strength – hardenability</td></tr><tr><td>Nickel</td><td>Toughness – strength – hardenability</td></tr><tr><td>Phosphorus</td><td>Strength but Temper embrittlement</td></tr><tr><td>Silicon</td><td>Deoxidation – hardenability</td></tr><tr><td>Sulfur</td><td>Machinability</td></tr><tr><td>Tellurium</td><td>Machinability</td></tr><tr><td>Titanium</td><td>Elimination of carbide precipitation</td></tr><tr><td>Vanadium</td><td>Fine grain - toughness</td></tr></table> | Alloying Element | Effect on Steel | Aluminum | Deoxidation – ease of nitriding | Boron 2-10 ppm, How added (effective Boron) | Hardenability | Carbon | Hardness – strength – wear | Chromium | Corrosion resistance - hardenability | Cobalt aka | Hardness – wear Reduces hardenability | Columbium (Niobium) | Elimination of carbide precipitation | Copper | Corrosion resistance – strength | Lead Not used anymore | Machinability | Manganese | Strength – hardenability – more response to heat treatment | Molybdenum | High temperature strength – hardenability | Nickel | Toughness – strength – hardenability | Phosphorus | Strength but Temper embrittlement | Silicon | Deoxidation – hardenability | Sulfur | Machinability | Tellurium | Machinability | Titanium | Elimination of carbide precipitation | Vanadium | Fine grain - toughness | 7 | 7 | 7 |
|---------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|-----------------|----------|---------------------------------|---------------------------------------------|---------------|--------|----------------------------|----------|--------------------------------------|------------|---------------------------------------|---------------------|--------------------------------------|--------|---------------------------------|-----------------------|---------------|-----------|------------------------------------------------------------|------------|-------------------------------------------|--------|--------------------------------------|------------|-----------------------------------|---------|-----------------------------|--------|---------------|-----------|---------------|----------|--------------------------------------|----------|------------------------|---|---|---|
| Alloying Element                            | Effect on Steel                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Aluminum                                    | Deoxidation – ease of nitriding                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Boron 2-10 ppm, How added (effective Boron) | Hardenability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Carbon                                      | Hardness – strength – wear                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Chromium                                    | Corrosion resistance - hardenability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Cobalt aka                                  | Hardness – wear Reduces hardenability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Columbium (Niobium)                         | Elimination of carbide precipitation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Copper                                      | Corrosion resistance – strength                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Lead Not used anymore                       | Machinability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Manganese                                   | Strength – hardenability – more response to heat treatment                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Molybdenum                                  | High temperature strength – hardenability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Nickel                                      | Toughness – strength – hardenability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Phosphorus                                  | Strength but Temper embrittlement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Silicon                                     | Deoxidation – hardenability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Sulfur                                      | Machinability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Tellurium                                   | Machinability                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Titanium                                    | Elimination of carbide precipitation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| Vanadium                                    | Fine grain - toughness                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| 3.                                          | <p>The advantages and disadvantages of gaseous fuels</p> <p>Advantages</p> <p>(a) They are clean in use.</p> <p>(b) They do not require any special burner.</p> <p>(c) They burn without any shoot, or smoke and ashes.</p> <p>(d) They are free from impurities found in solid and liquid fuels.</p> <p>Disadvantages</p> <p>(a) Very large storage tanks are needed.</p> <p>(b) They are highly inflammable, so chances of fire hazards in their use is high.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 7                | 7               | 7        |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
|                                             | or                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
| 4.                                          | <ul style="list-style-type: none"><li>• <b>Calorific value of a fuel</b> is the total quantity of heat liberated when one unit of fuel is burnt completely.</li><li>• <b>Gross or Higher calorific value (G.C.V or H.C.V):</b> It is the total quantity of heat liberated when one unit of fuel is burnt completely and products of combustion has been cooled to room temperature.</li><li>• <b>Net or Lower calorific value (N.C.V or L.C.V) :</b> It is the total quantity of heat liberated when one unit of fuel is burnt completely and products of combustion has been permitted to escape.<br/><math display="block">L.C.V = H.C.V - \text{Latent heat of water vapour formed}</math><math display="block">\text{Latent heat of steam} = 587 \text{ cal/gm}</math></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 7                | 7               | 7        |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |
|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                  |                 |          |                                 |                                             |               |        |                            |          |                                      |            |                                       |                     |                                      |        |                                 |                       |               |           |                                                            |            |                                           |        |                                      |            |                                   |         |                             |        |               |           |               |          |                                      |          |                        |   |   |   |

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                |   |   |   |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| 5. | <p style="text-align: center;"><b>Single-Acting Air Compressor Cylinder</b></p> <p style="text-align: center;">or</p>                                                                                                                                                                                                                                                                                                          | 7 | 7 | 7 |
| 6. | <p>The <b>Vapor Compression Refrigeration Cycle</b> involves <b>four components</b>: compressor, condenser, expansion valve/throttle valve and evaporator. It is a compression process, whose aim is to raise the refrigerant pressure, as it flows from an evaporator. The high-pressure refrigerant flows through a condenser/heat exchanger before attaining the initial low pressure and going back to the evaporator.</p> | 7 | 7 | 7 |
| 7. | <p style="text-align: center;"><b>Classification of refrigerants</b></p> <p style="text-align: center;">or</p>                                                                                                                                                                                                                                                                                                                 | 7 | 7 | 7 |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |   |   |   |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| 8.  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 7 | 7 | 7 |
| 9.  | <ol style="list-style-type: none"> <li>1. Infinite availability of the source</li> <li>2. Easy channeled and Temperature is flexible</li> <li>3. Safe and clean</li> <li>4. The transfer of power and the speed is very easy to set</li> <li>5. Can be stored and Easy utilized</li> <li>6. Low component cost &amp; light weight</li> <li>7. Pneumatic system can run continuously for long period</li> </ol>                                                                                                                                                                                                                     | 7 | 7 | 7 |
|     | or                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |   |   |   |
| 10. | <ul style="list-style-type: none"> <li>• Advantage: <ul style="list-style-type: none"> <li>– Able to generate extremely large forces from compact actuators</li> <li>– Easy to control speed</li> <li>– Easy to implement linear motion</li> </ul> </li> <li>• Disadvantage: <ul style="list-style-type: none"> <li>– Large infrastructure (high-pressure pump, tank, distribution lines)</li> <li>– Potential fluid leaks</li> <li>– Noisy operation</li> <li>– Vibration</li> <li>– Maintenance requirements, expensive</li> <li>– Characteristics of working fluids change with temperature and moisture</li> </ul> </li> </ul> | 7 | 7 | 7 |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                      |   |   |   |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|
| 11. | <ul style="list-style-type: none"> <li>• Possess low coefficient of friction.</li> <li>• Provide hard, wear resistant surface with a tough core.</li> <li>• Have high compressive strength.</li> <li>• Have high fatigue strength.</li> <li>• Be able to bear shocks and vibrations.</li> <li>• Possess high thermal conductivity to dissipate heat generated due to friction between the bearing and the rotating shaft.</li> </ul> | 7 | 7 | 7 |
|     | or                                                                                                                                                                                                                                                                                                                                                                                                                                   |   |   |   |
| 12  |                                                                                                                                                                                                                                                                                                                                                   | 7 | 7 | 7 |

# BLUE PRINT

## Mark Distribution

| Module | Hr/ Module                        | Marks/ Module<br>( $h_i / \sum H_i$ ) * 123 ( $\pm 5\%$ ) | Type of Questions |       |                  |       |                  |       |                  |       |
|--------|-----------------------------------|-----------------------------------------------------------|-------------------|-------|------------------|-------|------------------|-------|------------------|-------|
|        |                                   |                                                           | Part A            |       | Part B           |       | Part C           |       | Total            |       |
|        |                                   |                                                           | No. of Questions  | Marks | No. of Questions | Marks | No. of Questions | Marks | No. of Questions | Marks |
| 1      | 11                                | 31.5                                                      | 2                 | 2     | 1                | 3     | 4                | 28    | 7                | 33    |
| 2      | 12                                | 34.3                                                      | 3                 | 3     | 1                | 3     | 4                | 28    | 8                | 34    |
| 3      | 10                                | 28.6                                                      | 2                 | 2     | 4                | 12    | 2                | 14    | 8                | 28    |
| 4      | 10                                | 28.6                                                      | 2                 | 2     | 4                | 12    | 2                | 14    | 8                | 28    |
| Total  | 43(+2<br>for<br>internal<br>exam) | 123.0                                                     | 9                 | 9     | 10               | 30    | 12               | 84    | 31               | 123   |

## BLUE PRINT

### COGNITIVE LEVEL MARK DISTRIBUTION[Cognitive wise question analysis]

| Cognitive Level | Marks | % of Marks |
|-----------------|-------|------------|
| Remembering     | 42    | 34         |
| Understanding   | 81    | 66         |
| Applying        | 0     | 0          |
| Analysing       | 0     | 0          |
| Evaluating      | 0     | 0          |
| Creating        | 0     | 0          |
| Total           | 123   | 100        |



## QUESTION WISE ANALYSIS

**Course: Basic Mechanical Engineering Set 2**

| <b>Q. No.</b> | <b>Module Outcome</b> | <b>Cognitive Level</b> | <b>Marks</b> | <b>Time</b> |
|---------------|-----------------------|------------------------|--------------|-------------|
| <b>I.1</b>    | <b>M1.04</b>          | <b>Remembering</b>     | <b>1</b>     | <b>2</b>    |
| <b>I.2</b>    | <b>M1.02</b>          | <b>Remembering</b>     | <b>1</b>     | <b>2</b>    |
| <b>I.3</b>    | <b>M2.03</b>          | <b>Remembering</b>     | <b>1</b>     | <b>2</b>    |
| <b>I.4</b>    | <b>M2.04</b>          | <b>Remembering</b>     | <b>1</b>     | <b>2</b>    |
| <b>I.5</b>    | <b>M3.02</b>          | <b>Remembering</b>     | <b>1</b>     | <b>2</b>    |
| <b>I.6</b>    | <b>M3.04</b>          | <b>Remembering</b>     | <b>1</b>     | <b>2</b>    |
| <b>I.7</b>    | <b>M3.04</b>          | <b>Remembering</b>     | <b>1</b>     | <b>2</b>    |
| <b>I.8</b>    | <b>M4.02</b>          | <b>Remembering</b>     | <b>1</b>     | <b>2</b>    |
| <b>I.9</b>    | <b>M4.04</b>          | <b>Remembering</b>     | <b>1</b>     | <b>2</b>    |
| <b>II.1</b>   | <b>M1.01</b>          | <b>Remembering</b>     | <b>3</b>     | <b>7.5</b>  |
| <b>II.2</b>   | <b>M2.04</b>          | <b>Understanding</b>   | <b>3</b>     | <b>7.5</b>  |
| <b>II.3</b>   | <b>M3.02</b>          | <b>Understanding</b>   | <b>3</b>     | <b>7.5</b>  |
| <b>II.4</b>   | <b>M3.02</b>          | <b>Understanding</b>   | <b>3</b>     | <b>7.5</b>  |
| <b>II.5</b>   | <b>M3.02</b>          | <b>Remembering</b>     | <b>3</b>     | <b>7.5</b>  |
| <b>II.6</b>   | <b>M3.03</b>          | <b>Remembering</b>     | <b>3</b>     | <b>7.5</b>  |
| <b>II.7</b>   | <b>M4.04</b>          | <b>Understanding</b>   | <b>3</b>     | <b>7.5</b>  |
| <b>II.8</b>   | <b>M4.04</b>          | <b>Remembering</b>     | <b>3</b>     | <b>7.5</b>  |
| <b>II.9</b>   | <b>M4.05</b>          | <b>Understanding</b>   | <b>3</b>     | <b>7.5</b>  |
| <b>II.10</b>  | <b>M4.02</b>          | <b>Understanding</b>   | <b>3</b>     | <b>7.5</b>  |
| <b>III</b>    | <b>M1.04</b>          | <b>Remembering</b>     | <b>7</b>     | <b>17</b>   |
| <b>IV</b>     | <b>M1.03</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>V</b>      | <b>M1.04</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>VI</b>     | <b>M1.02</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>VII</b>    | <b>M2.03</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>VIII</b>   | <b>M2.03</b>          | <b>Remembering</b>     | <b>7</b>     | <b>17</b>   |
| <b>IX</b>     | <b>M2.03</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>X</b>      | <b>M2.04</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>XI</b>     | <b>M3.04</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>XII</b>    | <b>M3.04</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>XIII</b>   | <b>M4.03</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |
| <b>XIV</b>    | <b>M4.03</b>          | <b>Understanding</b>   | <b>7</b>     | <b>17</b>   |